Manuscript title

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Abstract

Summarize the question, approach, main result, and conclusion.

Keywords: keyword

1 Introduction

> OUTLINE NOTE
> 研究課題を示し，先行研究との関係を整理する．

Introduce the research question and its motivation. Discuss related work, for example (Example 2025).

2 Methods

2.1 Velocity measurements

Define the measured velocity from the particle position $x(t)$:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t}. \tag{1}$$

Use Equation (1) to obtain the velocity samples illustrated in Figure 1.

2.2 Sample mean

For N velocity samples, compute the sample mean as

$$\overline{v} = \frac{1}{N}\sum_{i=1}^{N} v_i. \tag{2}$$

2.2.1 Computational procedure

Implement the sample mean with the following procedure.

Algorithm 1: Compute the sample mean.

```
function mean(v, N)
    s ← 0
    for i ← 1 to N do
        s ← s + v[i]
    end for
    return s / N
end function
```

Refer to Algorithm 1 when describing the computation.

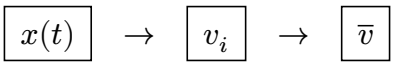


Figure 1: From position measurements to the mean velocity.

Table 1: Illustrative velocity measurements.

Sample	Velocity (m/s)
1	1.19
2	1.23
3	1.27

3 Results

3.1 Mean velocity

For the illustrative measurements in Table 1, Equation (2) gives the mean velocity. The mean and sample standard deviation are $(1.23 \pm 0.04)\,\mathrm{m\,s^{-1}}$. The latter is defined by Equation (3) in Appendix A.

TODO
- 測定値を実際の結果に置き換える.
- 既存研究との比較を追記する.

4 Conclusion

Summarize the findings and their implications.

Acknowledgments

Acknowledge helpful discussions, technical assistance, and funding support.

A Sample standard deviation

For the N measurements in Table 1, the sample standard deviation is

$$s_v = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(v_i - \overline{v})^2}. \tag{3}$$

References

Example A. 2025. A minimal research example. *Example Journal*. 1:1–5